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SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0720

COURSE OUTCOME COVERED

CO5:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL5)

Assessment Tool Weightage: 40%

82%

QUESTIONS: 3 **Total Marks:30**

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability

2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability

ANSWER:

A multinational company opening an office in a rural location faces the major constraint of **limited wired internet infrastructure**. Since cloud applications, video conferencing, and secure communication with headquarters are required, a reliable wireless-based internet solution should be selected.

A **4G/5G business broadband connection** can be used as the primary internet connection if adequate mobile coverage is available. If cellular coverage is weak or unavailable, **satellite internet** can be considered as an alternative. A secondary connection can be maintained as a backup to improve availability.

Recommended Network Design

```
      4G/5G or Satellite
            |
       Router / Firewall
            |
        Managed Switch
     _____/  |  \_____
    /        |        \
  PCs     Printer     Wi-Fi AP
   |
VPN to Headquarters
```

A business-grade router should support **VPN, firewall, QoS, and failover** features. A managed switch can be used to connect computers, printers, and other office devices. Wireless access points can provide connectivity to laptops and mobile devices.

A **VPN tunnel** should be established between the branch office and headquarters to encrypt sensitive business communication. Firewall rules should restrict unauthorized incoming traffic, while antivirus and endpoint security should be installed on employee devices.

For video conferencing and cloud applications, **Quality of Service (QoS)** can prioritize real-time traffic such as voice and video over less critical traffic. If the primary connection fails, the router can automatically switch to the backup connection.

Constraints Considered

- **Bandwidth:** QoS can prioritize video conferencing and business applications.
- **Reliability:** A secondary internet connection provides backup connectivity.
- **Cost:** 4G/5G is generally more affordable than dedicated leased-line infrastructure in remote locations.
- **Security:** VPN, firewall, encryption, and authentication protect company data.
- **Scalability:** Additional access points, switches, and users can be added later.

Conclusion

A **4G/5G-based connection with a backup link, business router/firewall, VPN, managed switch, and QoS** provides a practical balance between cost, reliability, security, and scalability for a rural branch office.

2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.

ANSWER:

A university with more than **3,000 students** requires a carefully designed wireless network because many users may connect simultaneously. Simply installing a small number of high-power access points would create congestion and interference. Therefore, access points should be strategically positioned based on building layout, user density, and expected traffic.

Recommended Wireless Design

Access points should be installed in areas with high user concentration, such as:

- Lecture halls
- Computer laboratories
- Libraries
- Hostels
- Cafeterias
- Common areas

High-density locations should receive more wireless capacity, while low-density areas can use fewer access points.

A simplified design is:

```
          Core Network
               |
        Wireless Controller
         /     |      \
       AP     AP       AP
      /        |        \
Classrooms   Library   Hostel
  / \          |        / \
Users Users  Users     Users
```

Frequency Band Selection

The **5 GHz band** should be preferred for high-performance connections because it provides more non-overlapping channels and generally experiences less congestion than 2.4 GHz.

Where supported, **6 GHz Wi-Fi** can provide additional spectrum and capacity for compatible modern devices.

The **2.4 GHz band** can be retained for devices that require greater range or do not support newer bands.

Automatic or centrally managed channel assignment can reduce interference between nearby access points.

Authentication and Security

For secure campus access, **WPA2-Enterprise or WPA3-Enterprise** with 802.1X authentication can be used. A RADIUS server can authenticate students and staff using their university credentials.

Separate networks can be created for:

- Students
- Faculty and staff
- Guests
- IoT devices

VLANs can isolate these networks and prevent unnecessary access between user groups.

Managing the Limited Number of Access Points

Since the university has a limited AP budget, access points should not simply be distributed equally across the campus. Instead, deployment should be based on **user density and traffic requirements**.

High-density areas should receive priority. AP transmit power can be adjusted to prevent excessive overlap, while band steering can encourage compatible devices to use higher-capacity frequency bands.

Constraints Considered

- **Coverage:** Strategic AP placement provides adequate coverage.
- **Performance:** High-density areas receive additional capacity.
- **Interference:** Channel planning and transmit-power control reduce interference.
- **Security:** Enterprise authentication protects user accounts and data.
- **Cost:** APs are prioritized according to user density instead of being installed everywhere.

Conclusion

A centrally managed, high-density wireless network using **5 GHz/6 GHz, strategic AP placement, enterprise authentication, VLAN segmentation, and intelligent channel planning** provides a good balance between coverage, performance, security, and the university's limited budget.

3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

ANSWER:

A financial institution handles highly sensitive customer and transaction information, so cybersecurity must be a major design priority. At the same time, the organization must control operational costs and avoid excessive performance degradation.

A **segmented network architecture** should be implemented so that sensitive systems are separated from ordinary employee devices.

Recommended Architecture

```
              INTERNET
                |
            Edge Firewall
                |
            DMZ / Public Zone
           /          \         \
       Web Server  VPN      Mail
                |
            Internal Firewall
                |
             Core Switch
        _________|__________
       /             |              \
     Users      Finance     Servers
    VLAN 10   VLAN 20    VLAN 30
```

Network Segmentation

Different departments and systems should be placed in separate **VLANs and IP subnets**.

For example:

Network	Example VLAN	Purpose
Employee Network	VLAN 10	Normal employee systems
Finance Network	VLAN 20	Financial operations
Server Network	VLAN 30	Internal servers
Guest Network	VLAN 40	Guest access
Management	VLAN 50	Network administration

Sensitive customer databases should be placed in a restricted server segment. Employees should only be allowed to access the resources required for their job.

Firewall Deployment

A firewall should be placed at the internet boundary to inspect incoming and outgoing traffic. Additional internal firewall controls can be used to protect sensitive server networks.

A **DMZ (Demilitarized Zone)** can host public-facing services such as web servers, keeping them separated from internal systems.

Firewall rules should follow the **least-privilege principle**, allowing only necessary traffic.

Authentication

Strong authentication should be implemented for employees and administrators. **Multi-factor authentication (MFA)** should be used for sensitive applications, VPN access, and administrative accounts.

Role-based access control can ensure that employees only access information required for their responsibilities.

Intrusion Detection and Prevention

An **IDS/IPS** can monitor network traffic and identify suspicious activities such as unauthorized access attempts, malware communication, and unusual traffic patterns.

Security logs should be centrally collected and monitored so that security teams can quickly investigate suspicious events.

Performance and Cost Considerations

Security mechanisms should be deployed strategically rather than inspecting every type of traffic unnecessarily. VLANs reduce unnecessary broadcast traffic, while firewall policies can be optimized to avoid excessive processing.

Open-source or existing enterprise security solutions can be considered where appropriate to control costs. Automated monitoring and centralized management can also reduce operational workload.

Constraints Considered

- **Security:** VLANs, firewalls, MFA, IDS/IPS, and access controls protect sensitive information.
- **Performance:** Network segmentation reduces unnecessary traffic and allows security controls to be strategically placed.
- **Compliance:** Logging, authentication, access control, and monitoring support regulatory requirements.
- **Budget:** Existing infrastructure and appropriately selected security solutions can reduce deployment costs.
- **Scalability:** New departments, applications, and security zones can be added without redesigning the complete network.

Conclusion

A **segmented and layered security architecture** combining VLANs, firewalls, DMZs, MFA, access controls, and IDS/IPS provides strong protection for a financial institution while maintaining acceptable network performance and controlling operational costs.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
A Understanding of Problem & Constraints	20% 2	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
B Problem Decomposition	15% 1.5	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
C Application of Constraints in Solution	20% 2	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
D Logical Reasoning & Strategy	15% 1.5	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
E Solution Accuracy & Feasibility	15% 1.5	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
F Creativity & Innovation	5% 0.5	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
G Clarity of Presentation	10% 1	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand

	A	B	C	D	E	F	G	
Q1	2	.5	1.5	1.5	1	0.5	1	8.5
Q2	1	1.5	2	1.5	1.5	0.5	0.5	8.5
Q3	1	1.5	1.5	1	1	0.5	1	7.5

24.5/30